

$$\chi^2 = 10.06$$

DATA
SET

DS1407

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence		
	Apartment	Duplex	Single Residence
1	8	20	1
2	16	8	9
3	10	10	14
4 or More	6	2	16
	40	40	40

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

$$RR: \chi^2_{0.01, 6} = 16.8119$$

$$\{ \chi^2 | \chi^2 > 16.8119 \}$$

$$36.498 > 16.8119 \Rightarrow \text{Reject } H_0$$

There is sufficient evidence to say there is a difference in family size distribution for the three types of residence

1 member family tends to live in duplex

2 members family tends to live in apartment

3 or more members family tends to live in single residence

H_0 : Family size distribution is independent of types of residence

H_a : Family size distribution depends on types of residence

$$\chi^2_{STAT} = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{0.01, (4-1)(3-1)} = \chi^2_{0.01, 6}$$

$$\hat{E}_{ij} = \frac{r_i \cdot c_j}{n}$$

	$\hat{E}_{ij}$	$\frac{(O_{ij} - \hat{E}_{ij})^2}{\hat{E}_{ij}}$
1	$\frac{29 \times 40}{120} = 9.667$	$\frac{(8 - 9.667)^2}{9.667} + \frac{(20 - 9.667)^2}{9.667} + \frac{(1 - 9.667)^2}{9.667}$
2	$\frac{33 \times 40}{120} = 11$	$\frac{(16 - 11)^2}{11} + \frac{(8 - 11)^2}{11} + \frac{(9 - 11)^2}{11}$
3	$\frac{34 \times 40}{120} = 11.33$	$\frac{(10 - 11.33)^2}{11.33} + \frac{(10 - 11.33)^2}{11.33} + \frac{(14 - 11.33)^2}{11.33}$
4 or more	$\frac{24 \times 40}{120} = 8$	$\frac{(6 - 8)^2}{8} + \frac{(2 - 8)^2}{8} + \frac{(16 - 8)^2}{8}$

$$\chi^2_{*} = 36.498$$